A Visually-Conditioned Reasoning Language Model for Diagnostic Workflow Reasoning and Pathology Report Generation from Whole-Slide Images

Shehla Mushtaq*

Submission to the MICCAI REG2026 Grand Challenge
(Reasoning and Explainable Grounding in Computational Pathology)

July 8, 2026

Abstract

Interpreting a gigapixel whole-slide image (WSI) is not a single act of classification but a *sequential reasoning process*: a pathologist confirms that tissue is present, identifies the dominant morphological pattern, grades and sub-types the lesion, and finally composes a structured report. The MICCAI REG2026 Grand Challenge operationalizes this process through two coupled tasks: *Workflow Reasoning*, which scores the recovered chain of diagnostic question–answer steps and the concluding report, and *Visual Grounding*, which tests whether that reasoning is anchored to visual evidence rather than produced by a memorized template. We formulate both tasks as visually-conditioned sequence generation and instantiate them with a single, parameter-efficient multimodal architecture. Each slide is tessellated and encoded by the CONCH pathology foundation model into a variable-length bag of 512-dimensional patch embeddings; a Perceiver resampler compresses this bag into a *fixed* set of K=128 latent visual tokens whose cost is decoupled from slide size; and a DeepSeek-R1-Distill-Qwen-1.5B reasoning decoder, adapted with Low-Rank Adaptation (LoRA), consumes these tokens and autoregressively emits an explicit chain of thought inside `<think>` tags followed by the structured report. The diagnostic workflow is serialized into the decoder's native chain-of-thought format, and training uses a standard language-modeling objective on the assistant tokens, so the model learns to reproduce the pathologist's stepwise reasoning and report end-to-end. Trained on 10,925 slide–reasoning pairs, the model attains a Workflow Reasoning Score of **0.721** on 223 held-out cases (Binary Path Validity 0.565, Edge-F1 0.807, Mean Edge Semantic Similarity 0.756, Final Report Score 0.655). We further contribute a CONCH-based region-of-interest grounding pipeline for the Visual Grounding metric and a fully offline, latency-hardened container that serves both challenge interfaces within strict per-case budgets. The results show that faithful and auditable diagnostic reasoning over gigapixel slides is attainable with a compact multimodal model trainable on two consumer GPUs.

Keywords: computational pathology, whole-slide images, multimodal large language models, chain-of-thought reasoning, report generation, multiple-instance learning, visual grounding, LoRA.

*Corresponding author: `shehlamushtaq63@gmail.com`

1 Introduction

Computational pathology is undergoing a shift from patch-level classification to slide-level, language-grounded interpretation. Two advances make this possible. First, self-supervised *vision foundation models* trained on tens of millions of histopathology tiles now yield patch embeddings that transfer across organs, stains, and tasks [3, 1]. Second, instruction-tuned *large language models* (LLMs) can ingest such embeddings as soft visual tokens and produce free-form clinical text [9, 11, 12]. Coupling the two promises systems that not only predict a diagnosis but articulate the reasoning behind it.

The obstacle is the nature of the input. A WSI is a gigapixel raster: at 20× magnification a single slide contains between 10^3 and 10^5 informative tiles, has no canonical reading order, and is annotated only at the slide level. Converting this unordered, arbitrarily large bag of tiles into a *diagnostically valid and auditable* reasoning trace—rather than a fluent but ungrounded narrative—is unsolved. Standard multiple-instance learning (MIL) pipelines [4, 2] collapse the bag into a single pooled vector optimized for a categorical label; they discard the intermediate evidence a report must reference and offer no mechanism for stepwise reasoning. Conversely, off-the-shelf multimodal LLMs are designed for a handful of natural-image tokens and neither scale to nor abstain over a bag of 10^5 patches.

The MICCAI **REG2026** Grand Challenge is constructed precisely around this gap. Instead of scoring a single label, it scores the *reasoning process*: the predicted chain of diagnostic questions and answers, the structure of that chain, the semantic correctness of each answer, and the quality of the final report; and, independently, whether the model's answers are *grounded*—sensitive to the tissue they describe and abstaining on non-informative background. Succeeding therefore requires a model that (i) consumes an arbitrarily large, orderless bag of patch features at bounded cost, (ii) reasons explicitly in a fixed, structured schema, and (iii) suppresses diagnostic claims on regions that carry no evidence. No prior architecture satisfies all three constraints simultaneously.

We address them with a single system built around three ideas. To bound cost independently of slide size, a Perceiver resampler cross-attends a fixed set of K=128 learned latent queries to the full patch bag, yielding a constant-size visual representation regardless of N. To make reasoning explicit and auditable, we retain the native <think> chain-of-thought behavior of a distilled R1 reasoning decoder and adapt it with LoRA rather than full fine-tuning, preserving its reasoning prior while training only 0.6% of the weights. To keep the reasoning explicit and consistent, we serialize the diagnostic workflow into the decoder's native chain-of-thought format—stepwise question/answer/next-question lines followed by the report—and train with a standard language-modeling loss on these assistant tokens, so the model learns to generate the entire workflow end-to-end.

Contributions. In summary, this work makes the following contributions. (1) We cast diagnostic workflow reasoning over WSIs as visually-conditioned sequence generation and present a compact architecture that couples a frozen CONCH encoder, a Perceiver resampler, and a LoRA-adapted DeepSeek-R1 reasoning decoder that natively thinks before it answers (Section 5). (2) We describe a target serialization that renders the diagnostic workflow in the decoder's native reasoning format, a masking scheme that supervises only the assistant tokens, and a report-preserving truncation policy that keeps the final report—the primary diagnostic deliverable—intact under a bounded token budget. (3) We report strong measured performance—a Workflow Reasoning Score of 0.721 on 223 held-out cases with a faithful reimplementation of all four metric components (Section 6). (4) We contribute a CONCH-based ROI grounding pipeline for the Visual Grounding metric and a fully offline, latency-hardened deployment that serves both challenge interfaces on modest hardware

(Sections 5.9 and 7).

2 Related Work

Weakly-supervised slide analysis. The dominant paradigm for slide-level prediction is attention-based MIL [4], refined by CLAM [2] into a practical pipeline that segments tissue, tessellates it into patches, and aggregates patch features through gated attention pooling. This family is highly effective for categorical endpoints but is *representationally lossy* for our setting: the bag is reduced to one pooled vector supervised by a single label, so the per-region evidence a report must cite is not retained, and the model has no capacity to emit an ordered reasoning chain. We keep CLAM only for its segmentation and tiling front end and replace the pooling head entirely with a resampler that preserves a distributed, query-indexed view of the bag.

Pathology vision foundation models. Self-supervised encoders such as UNI [3] and the vision–language model CONCH [1] produce transferable patch embeddings and have largely displaced ImageNet-pretrained backbones in histopathology. CONCH is additionally contrastively aligned to pathology text, which makes its patch space a natural interface to a language decoder; we therefore adopt frozen CONCH (ViT-B/16 [13]) patch features as the visual substrate and, crucially, reuse the identical encoder and normalization convention for region-of-interest queries, so a single ROI is numerically a one-patch slide. This encoder sharing is what lets one trained model serve both challenge tasks.

Bridging vision encoders to LLMs. Connecting a frozen visual encoder to a frozen LLM through a small trainable module was established by Flamingo [9], which introduced the Perceiver resampler [10]; by BLIP-2 [11], whose Q-Former plays the same role; and by LLaVA [12], which uses a linear projection. These bridges were designed for a small, fixed token budget from a single natural image. The WSI regime inverts their assumptions: the input is a bag of 10^3–10^5 tokens with no ordering. A linear projection would make decoder cost scale linearly with N and overwhelm the context window; a Q-Former's learned queries would face the same unbounded key set. We adopt the resampler formulation specifically because its latent queries cross-attend the entire bag while emitting a constant K tokens, moving the only N-dependent cost into a single linear-in-N cross-attention outside the decoder (Section 5.8).

Reasoning LLMs and parameter-efficient adaptation. Instruction-tuned decoders of the Qwen2 family [6], distilled into explicit chain-of-thought reasoners by DeepSeek-R1 [5], emit a delimited <think> reasoning span before their final answer. Prior pathology report-generation systems typically fine-tune a non-reasoning decoder to emit a report directly, discarding the intermediate rationale that REG2026 explicitly scores. We instead preserve the reasoning behavior and adapt the decoder with LoRA [7] (optionally QLoRA [8] with NF4 4-bit weights), training only low-rank adapters plus the resampler. This retains the decoder's reasoning prior—which a small pathology corpus could not re-learn from scratch—while keeping the trainable footprint small enough for two consumer GPUs. Generation quality is assessed with the classical lexical metrics BLEU [16] and ROUGE [17] together with biomedical embeddings from PubMedBERT [15], mirroring the official scoring.

Table 1: Dataset composition after preprocessing and feature matching. The test set is a 2% random hold-out (seed 42) used for all reported test metrics.

Quantity	Count
Processed supervised samples	11,220
Matched to a CONCH WSI feature file	11,148
Unmatched (missing WSI features, skipped)	72
Training samples	10,925
Held-out test samples	223

3 Problem Formulation and Data

3.1 The REG2026 task

A case supplies one WSI $\mathcal{S}$ and requires the model to reproduce the pathologist's *diagnostic workflow*: an ordered chain of T reasoning steps. Step t consists of a diagnostic *question* q_t, its *answer* a_t, and a pointer to the *next question* q_{t+1}, drawn from a canonical workflow vocabulary; the terminal step requests the final pathology report r. We treat the workflow as a single structured text output: the ordered steps and the report are serialized into one token sequence $y = (y_1, \ldots, y_M)$ (Section 5.5), and the model generates it autoregressively conditioned on the slide,

$$p_\theta(y \mid \mathcal{S}) = \prod_{m=1}^{M} p_\theta(y_m \mid y_{<m}, \mathcal{S})\,. \tag{1}$$

Performance is measured by two independent metrics—Workflow Reasoning (Metric A) and Visual Grounding (Metric B)—defined in Section 4; Metric A parses the generated sequence back into its constituent question/answer steps for scoring.

3.2 Corpus and annotations

Training uses the internal REG2026 chain-of-thought corpus (`train_CoT.json`). Each record carries a slide identifier and a list of reasoning steps under heterogeneous field names (e.g., `chain-of-thought`, `reasoning_steps`), which we normalize and serialize into one supervised `(prompt, target)` pair per case. Table 1 summarizes the corpus after matching each case to its precomputed CONCH feature file; 72 records whose WSI features were unavailable are skipped. The remainder is partitioned by a 2% random hold-out (seed 42) into 10,925 training and 223 test cases, the latter used for all reported test metrics.

3.3 Input representation

The reasoning model never sees raw pixels. Each slide is represented as a matrix $\mathbf{X} \in \mathbb{R}^{N\times d_v}$ of N tissue-patch embeddings of dimension d_v=512, where N varies from case to case. Following CLAM [2], tissue is segmented and tessellated into non-overlapping patches, and each patch p_i is encoded by CONCH,

$$x_i = f_{\mathrm{CONCH}}(p_i) \in \mathbb{R}^{d_v}, \qquad \mathbf{X} = [\, x_1, \ldots, x_N \,]^\top, \tag{2}$$

with contrastive projection and encoder-side normalization disabled so that raw embeddings are cached on disk. Features are L_2-normalized at load time, $\hat{x}_i = x_i/\|x_i\|_2$, which fixes their scale and makes ROIs encoded later by the same encoder numerically interchangeable with WSI patches—a property we exploit for Visual Grounding (Section 5.9).

4 Evaluation Metrics

We reimplement both official metrics faithfully, so that training-time model selection optimizes exactly what the leaderboard measures. Component weights are listed in Table 2.

4.1 Metric A: Workflow Reasoning

After canonicalization (lowercasing, whitespace collapse, trailing-punctuation removal), each reasoning step defines a directed edge $e = (\text{question}, \text{next\_question})$. Let E_{gt} and E_{pred} be the ground-truth and predicted edge sets. Metric A combines four terms.
Binary Path Validity (BPV) is an exact-match indicator on the reasoning graph,

$$\text{BPV} = \mathbf{1}[E_{\text{pred}} = E_{\text{gt}}]\,, \tag{3}$$

and is the strictest term: it rewards a case only when the *entire* edge set is recovered.
Edge-F1 awards partial structural credit. With $\text{TP} = |E_{\text{pred}} \cap E_{\text{gt}}|$, $\text{FP} = |E_{\text{pred}} \setminus E_{\text{gt}}|$, and $\text{FN} = |E_{\text{gt}} \setminus E_{\text{pred}}|$,

$$\text{Edge-F1} = \frac{2\,\text{TP}}{2\,\text{TP} + \text{FP} + \text{FN}}. \tag{4}$$

Mean Edge Semantic Similarity (MESS) measures answer correctness on expected non-final edges. With a biomedical text embedding $\phi(\cdot)$ and answers $a_e^{\text{gt}}, a_e^{\text{pred}}$,

$$s(e) = \begin{cases} \cos\big(\phi(a_e^{\text{gt}}),\, \phi(a_e^{\text{pred}})\big), & e \in E_{\text{pred}}, \\ 0, & e \notin E_{\text{pred}}, \end{cases} \qquad \text{MESS} = \frac{1}{|E_{\text{nonfinal}}|} \sum_{e \in E_{\text{nonfinal}}} s(e). \tag{5}$$

Final Report Score (FRS) grades the concluding report r^{pred} against r^{gt}, combining lexical overlap, keyword Jaccard, and a rescaled embedding cosine $\tilde{c} = \text{clip}((c - 0.5)/0.5,\, 0,\, 1)$:

$$\text{FRS} = 0.15\,(\text{ROUGE-L}_{\text{F1}} + \text{BLEU-4}) + 0.40\,\text{Jaccard}_{\text{kw}} + 0.30\,\tilde{c}. \tag{6}$$

The per-case Workflow Reasoning Score aggregates the four terms,

$$\text{WRS} = 0.05\,\text{BPV} + 0.30\,\text{Edge-F1} + 0.25\,\text{MESS} + 0.40\,\text{FRS}, \tag{7}$$

and the leaderboard value is the mean of WRS over all cases. Embeddings $\phi(\cdot)$ are mean-pooled PubMedBERT [15] vectors, with a deterministic hashed bag-of-keywords fallback that keeps the metric runnable fully offline. The 0.40 weight on FRS makes report quality the dominant contributor to the aggregate—each 0.01 of report quality is worth as much as 0.08 of BPV—which is why we take care to preserve the full report (Section 5.5) and treat report quality as the main avenue for improvement (Section 8).

4.2 Metric B: Visual Grounding

Metric B tests whether reasoning is tied to pixels. For each case the evaluator samples diagnostically relevant *tissue* ROIs T and non-tissue *background* ROIs B, and poses a small set Q of ROI-level questions; a pathology *judge* LLM grades each answer. Let $A(\rho, q)$ denote the model's answer

Table 2: Component weights of the two challenge metrics.

Metric	Component	Weight
A: Workflow Reasoning	Binary Path Validity (exact graph match)	0.05
	Edge-F1 (partial structural overlap)	0.30
	MESS (answer semantic similarity)	0.25
	Final Report Score	0.40
B: Visual Grounding	B1: Background Rejection	0.30
	B2: Input Sensitivity	0.30
	B3: Cross-region Consistency	0.40

to question q on ROI ρ, let $S(\cdot,\cdot) \in \{0,1\}$ be the judged clinical equivalence of two answers, let $J_{\mathrm{bg}} \in \{0,1\}$ be a background-rejection indicator, and let $\tilde{t}$ be a patch-masked perturbation of tissue ROI t. The three sub-scores are

$$\mathrm{B1} = \frac{1}{|B||Q|} \sum_{b\in B} \sum_{q\in Q} J_{\mathrm{bg}}(b,q), \tag{8}$$

$$\mathrm{B2} = \frac{1}{|T||Q|} \sum_{t\in T} \sum_{q\in Q} S\big(A(t,q), A(\tilde{t},q)\big), \tag{9}$$

$$\mathrm{B3} = \frac{1}{|T||B||Q|} \sum_{t\in T} \sum_{b\in B} \sum_{q\in Q} \big[\,1 - S\big(A(t,q), A(b,q)\big)\big], \tag{10}$$

so that B1 rewards abstention on background, B2 rewards sensitivity to perturbation, and B3 rewards separation between tissue and background answers, with $\mathrm{VG} = 0.30\,\mathrm{B1} + 0.30\,\mathrm{B2} + 0.40\,\mathrm{B3}$.

5 Method

5.1 Overview

Our system realizes (1) as a three-stage forward pass, shown in Figure 1. A frozen CONCH encoder maps a slide to a patch bag $\mathbf{X} \in \mathbb{R}^{N\times d_v}$ (Section 3.3). A trainable Perceiver resampler compresses this variable-length bag into a fixed set of K=128 latent visual tokens $\mathbf{V} \in \mathbb{R}^{K\times d}$ that live in the decoder's embedding space (Section 5.2). These tokens are prepended to the tokenized chat prompt and passed to a LoRA-adapted DeepSeek-R1 reasoning decoder, which autoregressively generates the <think> reasoning chain and the structured report (Sections 5.3–5.4). Only the resampler and the LoRA adapters are trained; CONCH and the decoder base weights remain frozen. The remainder of this section specifies each stage, the training objective (Section 5.6), inference (Section 5.7), and complexity (Section 5.8).

5.2 Perceiver resampler

The resampler is the module that reconciles an orderless, arbitrarily large bag with a decoder that expects a small, fixed number of tokens. An input projection $W_{\mathrm{in}} \in \mathbb{R}^{d_v\times d}$ lifts the normalized bag to the decoder width, $\mathbf{H} = \hat{\mathbf{X}}\,W_{\mathrm{in}} \in \mathbb{R}^{N\times d}$. A set of K learned latent queries $\mathbf{Z}^0 \in \mathbb{R}^{K\times d}$, shared across all slides, is refined over L=3 pre-norm blocks. Writing multi-head attention as

$$\mathrm{Attn}(\mathbf{q},\mathbf{k},\mathbf{v}) = \mathrm{softmax}\Big(\frac{\mathbf{q}\mathbf{k}^\top}{\sqrt{d_h}}\Big)\mathbf{v}, \qquad \mathrm{MHA}(\cdot) = [\mathrm{Attn}_1;\ldots;\mathrm{Attn}_h]W_O, \tag{11}$$

with h=8 heads of width d_h=d/h [14], block ℓ applies latent-to-patch cross-attention, latent self-attention, and a feed-forward network, each residual:

$$\mathbf{Z}'_\ell = \mathbf{Z}_{\ell-1} + \mathrm{MHA}\big(\mathrm{LN}(\mathbf{Z}_{\ell-1}),\, \mathbf{H},\, \mathbf{H}\big), \tag{12}$$
$$\mathbf{Z}''_\ell = \mathbf{Z}'_\ell + \mathrm{MHA}\big(\mathrm{LN}(\mathbf{Z}'_\ell),\, \mathrm{LN}(\mathbf{Z}'_\ell),\, \mathrm{LN}(\mathbf{Z}'_\ell)\big), \tag{13}$$
$$\mathbf{Z}_\ell = \mathbf{Z}''_\ell + \mathrm{FFN}\big(\mathrm{LN}(\mathbf{Z}''_\ell)\big). \tag{14}$$

The design is deliberate. Cross-attention (12) keys and values are the *patches themselves*, so at every layer the latents re-query the raw bag rather than a single pooled summary—this is what preserves the distributed, region-indexed evidence that pooling-based MIL discards. Self-attention (13) lets the K latents specialize and de-duplicate, allocating queries across distinct morphological regions. The output $\mathbf{V} := \mathbf{Z}_L \in \mathbb{R}^{K\times d}$ is constant in size for any N, so a slide with 10^2 or 10^5 patches presents the decoder with the same 128 tokens. The resampler runs in the decoder's bf16 compute dtype to bound the activation footprint of attending K queries over thousands of patches.

5.3 Multimodal fusion and prompt assembly

Fusion is performed in the decoder's input-embedding space rather than through a separate cross-attention stack, which keeps the decoder architecturally unmodified and lets LoRA adapt it in place. We build the chat prompt from a pathologist *system* message and a *user* instruction using the decoder's chat template, and embed its token ids with the decoder embedding matrix E to obtain $\mathbf{P} = E(\text{prompt}) \in \mathbb{R}^{|P|\times d}$. The visual tokens are prepended, forming the fused input sequence

$$\mathbf{U} = \big[\,\mathbf{V}\,;\,\mathbf{P}\,\big] \in \mathbb{R}^{(K+|P|)\times d}, \tag{15}$$

which is passed to the decoder as `inputs_embeds`. Because the visual and text streams share one embedding space and one causal self-attention stack, every generated token attends to all K visual tokens at every layer, so grounding is available throughout decoding rather than only at a bottleneck. The system prompt fixes the output schema and instructs the decoder to reuse the canonical workflow question strings verbatim, which is essential for the exact-match structure that Metric A rewards.

5.4 Visually-conditioned reasoning decoder

The decoder is DeepSeek-R1-Distill-Qwen-1.5B [5], a decoder distilled to emit an explicit reasoning span before its answer. We preserve this behavior and adapt it with LoRA [7]: for each adapted projection $W_0 \in \mathbb{R}^{d_\text{out}\times d_\text{in}}$ the forward map becomes

$$W_0\mathbf{u} \;\longmapsto\; W_0\mathbf{u} + \tfrac{\alpha}{r}\, BA\,\mathbf{u}, \qquad A \in \mathbb{R}^{r\times d_\text{in}},\; B \in \mathbb{R}^{d_\text{out}\times r}, \tag{16}$$

with rank r=16, scaling α=32, and dropout 0.05, applied to all attention and MLP projections (`q/k/v/o_proj`, `gate/up/down_proj`); B is zero-initialized so training begins at the pretrained function. Only $\{A, B\}$ and the resampler are trainable—roughly 0.6% of the decoder's parameters—so the reasoning prior is retained rather than overwritten by a small pathology corpus. The R1 chat template forces an opening `<think>` token, which we also supply in the target; the forced token is stripped from the target to avoid duplication. The same architecture supports QLoRA (NF4 4-bit base [8]) and `device_map` sharding for 7B/14B decoders, but the deployed configuration is the bf16 1.5B model (Table 3).

5.5 Target serialization

We serialize the diagnostic workflow into the reasoning decoder's native format. The ordered steps are rendered as numbered `Question:` / `Answer:` / `Next Question:` lines and wrapped, together with the report, into the R1 layout

<think> *reasoning* </think> Pathology Report: *report.*

This is precisely the layout the decoder was distilled to produce, so generation stays on the decoder's reasoning manifold rather than fighting it: the model writes its chain of thought inside <think> tags and then commits to the report after the marker. The format is line-oriented and unambiguous, which also lets the generated string be read back into its constituent steps without any post-processing. When the serialized target exceeds the 2048-token budget, truncation removes tokens from the *reasoning* head while preserving the entire final report, since the report is the primary diagnostic deliverable and must be emitted in full, whereas an over-length reasoning tail can be shortened with little loss of content.

5.6 Training objective

Each training sequence is $\mathbf{U} \parallel E(y)$, the fused input followed by the embedded target tokens $y = (y_1, \ldots, y_M)$. We minimize the negative log-likelihood of the target under teacher forcing, but supervise *only* the assistant span: labels are set to the ignore index over the visual and prompt positions, so the loss is

$$\mathcal{L}(\theta) = -\sum_{m=1}^{M} \mathbf{1}[m \in \mathcal{A}] \log p_\theta(y_m \mid y_{<m}, \mathbf{V}, \mathbf{P}), \tag{17}$$

where $\mathcal{A}$ indexes the assistant (reasoning-plus-report) tokens and θ collects the resampler and LoRA parameters. Masking the prompt and visual spans prevents the model from wasting capacity re-predicting its own inputs and concentrates the gradient on the reasoning steps and the report that constitute the diagnostic output. We optimize (17) with AdamW (learning rate 2×10^{-5}, gradient clipping at 1.0) for 3 epochs at effective batch size 32 (per-device batch 1, gradient accumulation 16, 2 GPUs under DistributedDataParallel). Gradient checkpointing in DDP *static-graph* mode bounds activation memory for the long ($\approx$ 2.7k-token) sequences: checkpointing reruns the forward pass during the backward pass, and static-graph mode reconciles this with the gradient reducer. After every epoch the model is generated-and-scored on a held-out subset with the full Metric A implementation, and a checkpoint (LoRA adapter, resampler, tokenizer, config) is saved. Hyperparameters are summarized in Table 3.

5.7 Inference

At test time the decoder receives only the fused input $\mathbf{U} = [\mathbf{V}; \mathbf{P}]$ and decodes greedily (arg max at each step) for up to 1800 new tokens. Because `inputs_embeds` is supplied, a decoder-only model returns only the newly generated tokens, so no slicing of the prompt is required; the generated string is passed directly through the Metric A parser. The same generator, given a one-patch "slide," answers ROI questions for Visual Grounding (Section 5.9).

5.8 Complexity

The design isolates all slide-size dependence in the resampler. Cross-attention in each of the L blocks costs $O(KNd)$ and self-attention/FFN cost $O(K^2d)$, giving a resampler cost of $O\big(L(KN + K^2)d\big)$

Table 3: Deployed model and training configuration.

Component	Setting
Patch encoder	CONCH ViT-B/16, frozen, 512-d output
Resampler	Perceiver, K=128 tokens, depth 3, 8 heads
Decoder (LLM)	DeepSeek-R1-Distill-Qwen-1.5B, bf16
Adaptation	LoRA r=16, α=32, dropout 0.05, all attn+MLP
Visual feature dim	512 (variant: 1024-d, 7B decoder, 256 tokens)
Max prompt / target len	512 / 2048 tokens
Optimizer	AdamW, lr 2×10^{-5}, grad-clip 1.0
Schedule	3 epochs; batch 1× accum 16× 2 GPUs (eff. 32)
Parallelism	DDP + gradient checkpointing + static graph
Precision	bf16 (LoRA + resampler trainable; base frozen)

that is *linear* in the patch count N. The decoder then operates on a sequence of length $K+|P|+|y|$ that is *independent* of N, so the entire language stack—by far the most expensive component—has cost invariant to slide size. This is the property that makes gigapixel inference tractable: doubling the number of patches at most doubles a single linear-in-N cross-attention, and never enlarges the decoder's context. The patch cap deployed for latency (Section 7) bounds even this term without changing the K=128 tokens the decoder sees.

5.9 Visual Grounding pipeline

Metric B is served by the same trained model through a self-contained two-stage pipeline, kept separate from the training code (it imports only `load_generator` and `load_wsi_features`). *Stage 1* encodes each 256×256 ROI with a vendored, pure-Python CONCH into the identical $[1, 512]$ format as a WSI patch, exploiting the encoder- and normalization-sharing of Section 3.3 so that an ROI is numerically a one-patch slide. *Stage 2* queries the trained reasoning model per ROI and records each answer together with its tissue/background label, its original/perturbed variant, and the B3 pairing metadata required to compute B1/B2/B3. An optional ROI-focused system prompt biases the model toward brief, grounded answers and explicit abstention on non-informative regions, directly targeting the behavior B1 and B3 reward. The released ROI set contains 18 regions (12 tissue, 6 background; 12 original, 6 perturbed). Stage-1 features and Stage-2 answers run end-to-end; the final B1/B2/B3 aggregation requires an external pathology judge LLM and is the remaining step (Section 9).

6 Experiments

6.1 Implementation details

Patch extraction runs under PyTorch 2.1 with timm and the CONCH weights; checkpoint loading and generation use a newer PyTorch 2.8 / Transformers 4.57 / PEFT 0.17 stack so the LoRA adapter (serialized with a recent PEFT) deserializes. Generation is greedy (`do_sample=False`) with up to 1800 new tokens. Metric A uses PubMedBERT embeddings for MESS and the report cosine. All test numbers are computed on the 223-case hold-out of Table 1.

Table 4: Workflow Reasoning results on the 223-case held-out test set. All components lie in $[0,1]$; higher is better. Parenthetical values are the Metric A weights of Eq. (7).

Component (weight)	Score
Binary Path Validity (0.05)	0.565
Edge-F1 (0.30)	0.807
MESS (0.25)	0.756
Final Report Score (0.40)	0.655
Workflow Reasoning Score	**0.721**

6.2 Main results

Table 4 reports the four Metric A components and the combined score. The model attains a Workflow Reasoning Score of **0.721**. The profile of the four terms is internally coherent and interpretable. Edge-F1 is the strongest component at 0.807, indicating that the predicted reasoning steps overlap the reference steps well and that the model reliably reproduces the canonical question/answer transitions. Exact-match BPV (0.565) is, as expected, the hardest term, since it credits a case only when *every* transition is recovered; the gap between Edge-F1 and BPV is precisely the population of cases in which most—but not all—transitions are correct. MESS (0.756) shows that answers on recovered edges are semantically close to the references, confirming the model reasons about content rather than merely reproducing question strings. The Final Report Score (0.655), the most heavily weighted term, indicates clinically plausible reports and, given its 0.40 weight, is the single largest contributor to the aggregate.

6.3 Report-generation quality

The Final Report Score in Table 4 is a challenge-specific composite; to place the generated reports on a footing comparable to the report-generation literature, Table 5 reports standard surface- and embedding-level metrics computed on the same 223 cases, comparing the *extracted* pathology report against its reference. BLEU-n and GLEU [18] are corpus-level (NLTK, method-1 smoothing), ROUGE-L is the mean per-report stemmed F1, and BERTScore [19] is the mean token-level F1 under a RoBERTa-large backbone (no baseline rescaling, so absolute values are high, as is typical for this metric).

The results are consistent with the qualitative picture from FRS. Lexical overlap is high at the unigram level (BLEU-1 0.725, ROUGE-L 0.736) and decays gracefully with n-gram order (BLEU-2/3/4 = 0.676/0.634/0.596), the signature of reports that reproduce the correct clinical vocabulary and short phrases while diverging on exact long-span wording. GLEU (0.540), a recall-sensitive corpus metric, tracks BLEU-4 closely. The embedding-level BERTScore F1 (0.959) confirms that generated and reference reports are close in a contextual semantic space, not merely in surface tokens. We stress that the BLEU-4 reported here is a corpus-level, smoothed score on the report text and is *not* the same quantity as the BLEU-4 sub-term folded into FRS (Section 4), which uses a different tokenizer and enters only at weight 0.15; the two numbers should not be compared directly.

6.4 Training dynamics and component ablation over epochs

Because every component is scored after each epoch, the training trajectory doubles as a controlled study of *which* capabilities the resampler and adapters acquire, and in what order. Table 6 tracks the

Table 5: Report-generation quality on the 223-case test set, comparing the generated pathology report against the reference. BLEU-n and GLEU are corpus-level with method-1 smoothing; ROUGE-L is the mean per-report stemmed F1; BERTScore is the mean token-level F1 (RoBERTa-large, no baseline rescaling). All scores lie in $[0, 1]$; higher is better.

Metric	Score
BLEU-1	0.725
BLEU-2	0.676
BLEU-3	0.634
BLEU-4	0.596
ROUGE-L (F1)	0.736
GLEU	0.540
BERTScore (F1)	0.959

Table 6: In-training validation (random 10-case subset) by epoch. Values are optimistic and noisier than Table 4 due to the small resampled subset; the trends, not the absolute levels, are the object of study.

Component	Epoch 1	Epoch 2	Epoch 3
Binary Path Validity	0.300	0.500	0.700
Edge-F1	0.666	0.737	0.889
MESS	0.590	0.693	0.889
Final Report Score	0.453	0.563	0.775
Workflow Reasoning Score	0.543	0.645	**0.834**

in-training validation score on a random 10-case subset; these values are optimistic and noisier than the full test set due to the small resampled subset, and we read them for trends rather than absolute levels. Every component improves monotonically across the three epochs, and the combined score rises from 0.543 to 0.834. The most informative signal is the differential rate of improvement: BPV more than doubles ($0.300 \to 0.700$) while Edge-F1 is already high after epoch 1 (0.666). This ordering—partial structural overlap emerging early, exact match consolidating late—is what one expects if the model first learns the canonical question vocabulary and local transitions and only later stitches them into globally consistent chains.[1] FRS improves in lockstep ($0.453 \to 0.775$), indicating that reasoning structure and report quality are learned jointly rather than traded off, consistent with the shared assistant-token objective of (17).

6.5 Qualitative analysis of Visual Grounding

The ROI extraction and question-answering stages run end-to-end on the released 18-ROI set, producing per-ROI answers with the metadata required for B1/B2/B3. Qualitatively, with the ROI-focused system prompt the model abstains on background regions (e.g., "this region is background / not assessable") while giving content-bearing answers on tissue ROIs—the behavior B1 (background rejection) and B3 (tissue–background separation) reward. This asymmetry is direct evidence that the model conditions on the visual token rather than emitting a context-independent template, since the prompt is held fixed across ROIs and only the single visual token changes. Quantitative

[1] Here "transition" refers to the metric's edge between a question and its next question; it is a property of how Metric A scores the sequence, not a structure the model predicts explicitly.

B1/B2/B3 values require the external judge model and are deferred (Section 9); we do not claim them here.

6.6 Computational efficiency

The efficiency of the system follows from two decisions analyzed above. First, the resampler makes decoder cost independent of slide size (Section 5.8), so the language stack processes the same $K+|P|+|y|$ tokens whether a slide has 10^2 or 10^5 patches. Second, parameter-efficient adaptation trains only $\sim 0.6\%$ of the decoder plus the resampler, which is what permits the effective-batch-32 regime of Table 3 on two consumer GPUs. At deployment, merging the LoRA adapter into the base weights removes the per-token low-rank matmuls at no change to outputs, and a patch cap bounds the sole linear-in-N term (Section 7).

6.7 Error analysis

The metric decomposition localizes the model's residual errors. The dominant signed gap is between Edge-F1 (0.807) and BPV (0.565): most failures are not wholesale hallucinations of the reasoning chain but single missing or spurious transitions that forfeit the all-or-nothing BPV credit while leaving Edge-F1 largely intact. Because BPV carries only weight 0.05, these near-misses cost little in aggregate, but they identify long or atypical chains—where one mis-predicted transition breaks exact match—as the structural failure mode. On content, MESS (0.756) exceeds FRS (0.655), indicating that short intermediate answers are recovered more reliably than the long free-text report; report generation, not edge prediction, is therefore the accuracy bottleneck, in agreement with the weighting analysis below.

6.8 Limitations

The evaluation has three notable limits. Metric B is not yet scored end-to-end: ROI features and answers are produced, but B1/B2/B3 require a pathology judge LLM that is not integrated. The per-epoch numbers of Table 6 come from a 10-case subset and are optimistic relative to the 223-case test set. Finally, all results use greedy decoding; sampling-based decoding and larger validation splits are unexamined. These points are expanded in Section 9.

7 Containerized Deployment

The challenge executes submissions as an offline container, with weights mounted separately and no network access. A single image serves both interfaces. **Interface 0** maps an ROI `.jpeg` to one answer string (CONCH $\rightarrow [1, 512] \rightarrow$ generation), and **Interface 1** maps a WSI `.tiff` to a chain-of-thought array (pyramid conversion if needed $\rightarrow$ CLAM segmentation and patching $\rightarrow$ CONCH $[N, 512] \rightarrow$ generation $\rightarrow$ schema parser). Each handler wraps inference in a guarded fallback so that no single case can leave a missing output.

A first online run completed most cases but halted on a single outlier slide that exceeded the per-case time limit. We added four latency safeguards, each chosen to leave accuracy on normal cases unchanged. (1) *Patch cap*: if a slide yields more than REG_MAX_PATCHES (12,000) patches, coordinates are uniformly (evenly-strided) subsampled before encoding; because the resampler compresses to 128 tokens regardless (Section 5.8), coverage is preserved and normal slides are untouched. (2) *LoRA merge*: the adapter is merged into the base weights at load time (`merge_and_unload`), making decoding mathematically identical while dropping the per-token low-rank matmuls. (3) *Throughput*: larger CONCH inference batches and more data-loader workers (pure I/O, no effect on outputs). (4) *Generation budget*: a `MaxTimeCriteria` (180 s default) returns truncated-but-parseable text instead of allowing a runaway decode to kill the submission. Because the merge happens at load time, the weights are unchanged and only the container image is re-uploaded.

8 Discussion

Three design choices explain the measured behavior. *Native-format supervision* (Section 5.5) serializes the workflow in the decoder's own reasoning layout and restricts the loss to assistant tokens (17), so training concentrates on the stepwise reasoning and the report; this is the mechanism behind the strong Edge-F1 and the steep BPV gains across epochs. *Constant-size visual grounding* (Sections 5.2 and 5.8) decouples decoder cost from slide size, which is what makes both training memory and worst-case latency manageable and, at deployment, what lets a patch cap bound runtime without altering the 128 tokens the decoder consumes. *Parameter efficiency* (Section 5.4) trains only the adapters and resampler over a frozen 1.5B reasoner, retaining its native reasoning ability on modest hardware.

The held-out profile also indicates where the next gains lie. Report quality is the binding constraint: at weight 0.40 in (7), each 0.01 of FRS is worth as much as 0.08 of BPV, so the FRS of 0.655—the lowest-scoring high-weight term—is the highest-value target. The error analysis reinforces this: intermediate answers (MESS 0.756) are recovered more reliably than the report (FRS 0.655), so report-focused decoding, retrieval of canonical report templates, or light reinforcement against FRS are the most promising directions.

9 Limitations and Future Work

Metric B is not yet scored end-to-end. ROI features and answers are produced, but B1/B2/B3 require a pathology judge LLM that is not yet integrated; wiring in a local judge and reporting Visual Grounding numbers is the immediate next step. *Report quality is the binding constraint*, and given its 0.40 weight, report-focused decoding, template retrieval, or reinforcement against the Final Report Score are the highest-value improvements. *Decoder scale* is capped for latency and memory: the codebase already supports 7B/14B decoders via QLoRA and `device_map`, and scaling is expected to raise MESS and report fidelity most. Finally, *decoding and validation* are conservative: results use greedy generation, and the per-epoch numbers come from a 10-case subset that is optimistic relative to the 223-case test set, so sampling-based decoding and larger validation splits deserve study.

10 Conclusion

We presented a compact, end-to-end system for diagnostic workflow reasoning and pathology report generation from whole-slide images. Casting the task as visually-conditioned sequence generation, we compress a frozen CONCH patch bag into a constant 128 visual tokens with a Perceiver resampler and let a LoRA-adapted DeepSeek-R1 reasoning decoder generate the entire diagnostic chain in its native reasoning format. Trained only on low-rank adapters and the resampler, the model reaches a Workflow Reasoning Score of 0.721 on a 223-case hold-out, recovering the reasoning structure well (Edge-F1 0.807). We additionally contribute a CONCH-based ROI grounding pipeline for the Visual Grounding metric and a fully offline, latency-hardened container serving both challenge interfaces. The system demonstrates that faithful and auditable pathology reasoning over gigapixel slides is achievable with a parameter-efficient multimodal model on modest hardware.

Reproducibility and Data Availability

The model, training, and evaluation code—feature extraction, Perceiver resampler, reasoning decoder, the Metric A reimplementation, the ROI grounding pipeline, and the deployment container—are organized in the project repository. Training uses the REG2026 chain-of-thought corpus; CONCH weights follow the CONCH release. Each checkpoint stores the LoRA adapter, resampler weights, tokenizer, and configuration for exact reload.

References

[1] M. Y. Lu, B. Chen, D. F. K. Williamson, *et al.*, "A visual–language foundation model for computational pathology," *Nature Medicine*, vol. 30, pp. 863–874, 2024.

[2] M. Y. Lu, D. F. K. Williamson, T. Y. Chen, *et al.*, "Data-efficient and weakly supervised computational pathology on whole-slide images," *Nature Biomedical Engineering*, vol. 5, pp. 555–570, 2021.

[3] R. J. Chen, T. Ding, M. Y. Lu, *et al.*, "Towards a general-purpose foundation model for computational pathology," *Nature Medicine*, vol. 30, pp. 850–862, 2024.

[4] M. Ilse, J. M. Tomczak, and M. Welling, "Attention-based deep multiple instance learning," in *Proc. ICML*, 2018, pp. 2127–2136.

[5] DeepSeek-AI, "DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning," arXiv:2501.12948, 2025.

[6] A. Yang, B. Yang, B. Hui, *et al.*, "Qwen2 technical report," arXiv:2407.10671, 2024.

[7] E. J. Hu, Y. Shen, P. Wallis, *et al.*, "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*, 2022.

[8] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in *Proc. NeurIPS*, 2023.

[9] J.-B. Alayrac, J. Donahue, P. Luc, *et al.*, "Flamingo: A visual language model for few-shot learning," in *Proc. NeurIPS*, 2022.

[10] A. Jaegle, F. Gimeno, A. Brock, *et al.*, "Perceiver: General perception with iterative attention," in *Proc. ICML*, 2021, pp. 4651–4664.

[11] J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. ICML*, 2023.

[12] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," in *Proc. NeurIPS*, 2023.

[13] A. Dosovitskiy, L. Beyer, A. Kolesnikov, *et al.*, "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. ICLR*, 2021.

[14] A. Vaswani, N. Shazeer, N. Parmar, *et al.*, "Attention is all you need," in *Proc. NeurIPS*, 2017.

[15] Y. Gu, R. Tinn, H. Cheng, *et al.*, "Domain-specific language model pretraining for biomedical natural language processing," *ACM Trans. Computing for Healthcare*, vol. 3, no. 1, 2021.

[16] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "BLEU: A method for automatic evaluation of machine translation," in *Proc. ACL*, 2002, pp. 311–318.

[17] C.-Y. Lin, "ROUGE: A package for automatic evaluation of summaries," in *Text Summarization Branches Out*, ACL Workshop, 2004, pp. 74–81.

[18] Y. Wu, M. Schuster, Z. Chen, *et al.*, "Google's neural machine translation system: Bridging the gap between human and machine translation," arXiv:1609.08144, 2016.

[19] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, "BERTScore: Evaluating text generation with BERT," in *Proc. ICLR*, 2020.

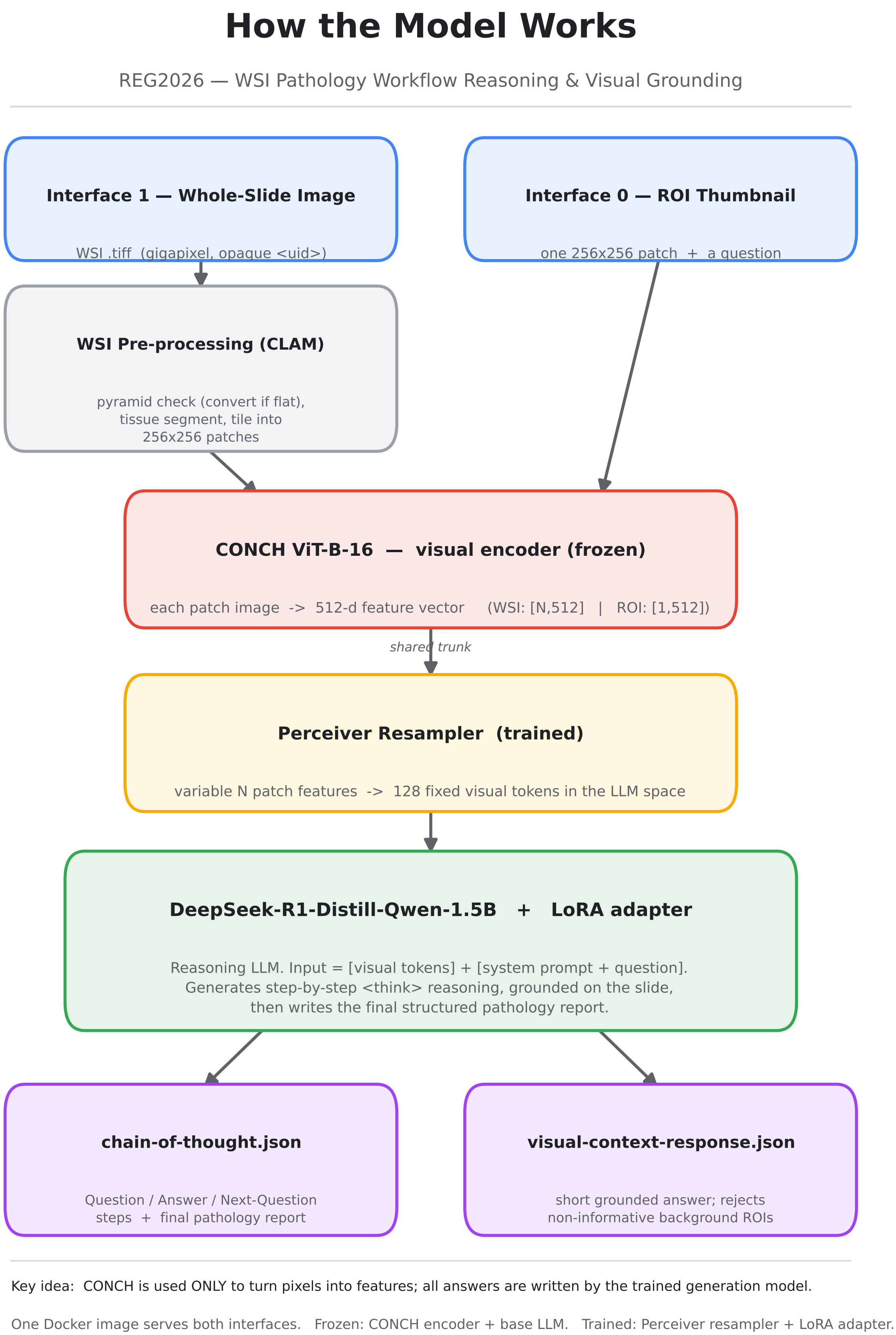

Figure 1: End-to-end architecture. CONCH (ViT-B/16) encodes tissue patches into a variable-length bag $\mathbf{X} \in \mathbb{R}^{N\times 512}$. A Perceiver resampler with K=128 learned latent queries cross-attends the bag at every layer and emits a fixed set of visual tokens $\mathbf{V}$. These are concatenated with the embedded system/user chat prompt and fed to a LoRA-adapted DeepSeek-R1-Distill-Qwen decoder, which generates the `<think>`